Technological University of the Philippines Visayas
Capitan Sabi Street, Barangay Zone 12, Talisay City, Negros Occidental
Email: tup@tup.edu.ph

# CPE3A2025 4-BIT CPU ARCHITECTURE

## DATA SHEET

Instruction Set • Description

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Technological University of the Philippines Visayas
Capitan Sabi Street, Barangay Zone 12, Talisay City, Negros Occidental
Email: tup@tup.edu.ph

# INSTRUCTION SET

| Mnemonic | Opcode | Operand | Description | 12-BIT Instruction | |
|---|---|---|---|---|---|
| | | | | OPCODE | OPERAND |
| **NOOP** | 0000 | NULL | CPU sleeps for 1 cycles | 0000 | NULL |
| **LOADW** | 0001 | 8-BIT Data Input | Load W from input | 0001 | bbbbbbbb |
| **MOVW** | 0010 | 8-BIT Literal | Load W from immediate data | 0010 | bbbbbbbb |
| **MOVWM** | 0011 | 5-BIT Address | Move data from W to memory | 0011 | 000bbbbb |
| **MOVMW** | 0100 | 5-BIT Address | Move data from memory to W | 0100 | 000bbbbb |
| **CMP** | 0101 | 8-BIT Literal | Compare W with immediate data | 0101 | bbbbbbbb |
| **SKIPE** | 0110 | NULL | Skip next instruction if ZF is 1 | 0110 | NULL |
| **JMP** | 0111 | 5-BIT Ins Address | Jumps to target instruction address | 0111 | 000bbbbb |
| **OUTW** | 1000 | NULL | Out W to output port | 1000 | NULL |
| **ADDW** | 1001 | 8-BIT Literal | Add immediate data to W | 1001 | bbbbbbbb |
| **ADDMW** | 1010 | 5-BIT Address | Add data from memory to W | 1010 | 000bbbbb |
| **SUBW** | 1011 | 8-BIT Literal | Subtract immediate data from W | 1011 | bbbbbbbb |
| **SUBMW** | 1100 | 5-BIT Address | Subtract data memory from W | 1100 | 000bbbbb |
| **ANDW** | 1101 | 8-BIT Literal | And W with immediate data | 1101 | bbbbbbbb |
| **ORW** | 1110 | 8-BIT Literal | Or W with immediate data | 1110 | bbbbbbbb |
| **NOTW** | 1111 | NULL | Not W | 1111 | NULL |

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# INSTRUCTION DESCRIPTION

## NOOP — No Operation

**Opcode:** 0000

**Operands:** None

**Operation:** CPU sleeps for 1 cycles

**Encoding:** 0000 00000000

**Description:** The CPU executes no-operation instruction and proceeds to the next instruction.

**Cycles:** 2

## LOADW — Load Literal to Working Register

**Opcode:** 0001

**Operands:** 8-BIT Data Input

**Operation:** Load W from input

**Encoding:** 0001 bbbbbbbb

**Description:** Loads an 8-bit value from an external input into the W register.

**Cycles:** 3

## MOVW — Move Literal to Working Register

**Opcode:** 0010

**Operands:** 8-BIT Literal

**Operation:** Load W from immediate data

**Encoding:** 0010 bbbbbbbb

**Description:** Loads an 8-bit immediate value into the W register.

**Cycles:** 3

## MOVWM Move Working Register to Memory

**Opcode:** 0011

**Operands:** 5-BIT Address

**Operation:** Move data from W to memory

**Encoding:** 0011 000bbbbb

**Description:** Moves the value from the W register to a specified 5-bit memory address.

**Cycles:** 3

## MOVMW Move Memory to Working Register

**Opcode:** 0100

**Operands:** 5-BIT Address

**Operation:** Move data from memory to W

**Encoding:** 0100 000bbbbb

**Description:** Moves data from a specified 5-bit memory address into the W register.

**Cycles:** 4

## CMP Compare

**Opcode:** 0101

**Operands:** 8-BIT Literal

**Operation:** Compare W with immediate data

**Encoding:** 0101 bbbbbbbb

**Description:** Compares the W register with an 8-bit literal and updates status flags.

**Cycles:** 4

## SKIPE — Skip if Equal

**Opcode:** 0110

**Operands:** None

**Operation:** Skip next instruction if ZF is 1

**Encoding:** 0110 00000000

**Description:** Skips the next instruction if the zero flag (ZF) is set.

**Cycles:** 2

## JMP — Jump

**Opcode:** 0111

**Operands:** 5-BIT Instruction Address

**Operation:** Jumps to target instruction address

**Encoding:** 0111 000bbbbb

**Description:** Jumps to a specific instruction address based on a 5-bit value.

**Cycles:** 2

## OUTW — Output Literal to Working Register

**Opcode:** 1000

**Operands:** None

**Operation:** Output W to the output port

**Encoding:** 1000 00000000

**Description:** Sends the contents of the W register to an output port.

**Cycles:** 2

## ADDW — Add Literal to Working Register

**Opcode:** 1001

**Operands:** 8-BIT Literal

**Operation:** Add immediate data to W

**Encoding:** 1001 bbbbbbbb

**Description:** Adds an 8-bit literal value to the W register and stores the result in W.

**Cycles:** 4

## ADDMW — Add Memory to Working Register

**Opcode:** 1010

**Operands:** 5-BIT Address

**Operation:** Add data from memory to W

**Encoding:** 1010 000bbbbb

**Description:** Adds the value at a specified 5-bit memory address to W and stores the result in W.

**Cycles:** 5

## SUBW — Subtract Literal from Working Register

**Opcode:** 1011

**Operands:** 8-BIT Literal

**Operation:** Subtract immediate data from W

**Encoding:** 1011 bbbbbbbb

**Description:** Subtracts an 8-bit literal value from the W register and stores the result in W.

**Cycles:** 4

## SUBMW — Subtract Memory from Working Register

| | |
|---|---|
| **Opcode:** | 1100 |
| **Operands:** | 5-BIT Address |
| **Operation:** | Subtract data memory from W |
| **Encoding:** | 1100 000bbbbb |
| **Description:** | Subtracts the value at a specified 5-bit memory address from W and stores the result in W. |
| **Cycles:** | 5 |

## ANDW — AND Literal with Working Register

| | |
|---|---|
| **Opcode:** | 1101 |
| **Operands:** | 8-BIT Literal |
| **Operation:** | And W with immediate data |
| **Encoding:** | 1101 bbbbbbbb |
| **Description:** | Performs a bitwise AND between W and an 8-bit literal, storing the result in W. |
| **Cycles:** | 4 |

## ORW — OR Literal with Working Register

| | |
|---|---|
| **Opcode:** | 1110 |
| **Operands:** | 8-BIT Literal |
| **Operation:** | Or W with immediate data |
| **Encoding:** | 1110 bbbbbbbb |
| **Description:** | Performs a bitwise OR between W and an 8-bit literal, storing the result in W. |
| **Cycles:** | 4 |

# NOTW NOT Literal to Working Register

| | |
|---|---|
| **Opcode:** | 1111 |
| **Operands:** | None |
| **Operation:** | Not W |
| **Encoding:** | 1111 bbbbbbbb |
| **Description:** | Performs a bitwise NOT operation on W and stores the result in W. |
| **Cycles:** | 4 |